



UnionPay Integrated Circuit Card Specifications —Auxiliary Specifications

Part V Debit/Credit Application Personalization Guide

Version 2017

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Summary of Revisions

The change listed below is associated with the current version.

Change Description	Revised/ Deleted/ Added	Location
Added Section 7.5 to describe the Common AID personalization requirements.	Added	7.5

1 Application Scope

This book applies to all UPI Participants.

2 Normative References

The following documents are required for applications of this document. For dated reference files, only the version on the specified date is applicable to this document. For undated reference files, the latest version (including all updating list) are applicable to this document.

EMV Card Personalization Specification V1.1

Global Platform Card Specification V2.1.1

ISO/IEC 7816-3:1997 Id cards - IC cards with contact point Part 3: Electronic signals and transmission protocol

ISO/IEC 9797-1:1999 Information technology - security technology - message authentication code - Part 1: Block encryption mechanism

ISO/IEC 10116:1997 Information technology - n bit block encryption algorithm calculation mode

ISO/IEC 14443-4:2008 Id cards - Contactless IC cards - Proximity card - Part 4: transmission protocol

ISO/IEC 18033-3:2005 Information technology - security technology - encryption algorithm - Part 3: Block encryption

3 Symbols and Acronyms

3.1 Symbols

3.1.1 Hexadecimal Symbols

Data represented in hexadecimal are all enclosed with single quotes, for example 27509 in decimal converted into hexadecimal is '6B75'.

3.1.2 Binary Symbols

Numerical value represented in binary are always followed with a lower case 'b', for example hexadecimal '08' represented in binary is 00001000b.

3.1.3 Compressed Numeric Symbols

If a compressed number contains an odd number of digits, then at a minimum a hexadecimal 'F' should be added at the end. For example, 456 in decimal represented as compressed data type is '456F', stored in a 2 bytes long.

3.1.4 Data Symbols

Supported formats include:

- n (number)
- cn (compressed numeric)
- b(binary)
- an(alphanumeric)
- ans(special alphanumeric)
- Var. (Var.iable)

When the defined data length exceeds actual data length, and positions are not all filled, padding rules are as follows:

- format n data elements right align, left pad with 0s
- format cn data elements left align, right pad with Fs
- format an data elements left align, right pad with 0s
- format ans data elements left align, right pad with 0s

3.2 Acronym

The following acronyms and symbol representations are applicable for this Specification

Acronym	Description
CBC	Cipher Block Chaining

Acronym	Description
CDA	Combined DDA/AC Generation
C-MAC	Command Message Authentication Code—MAC for secure communication during personalization command processing
CPM	Common Personalization Method
CPS	Card Personalization Specification
AAC	Application Authentication Cryptogram
ADA	Application Default Action
ADF	Application Definition File
AFL	Application File Locator
AID	Application Identifier
AIP	Application Interchange Profile
Application	An application resident in an IC payment card
Application Command	For this document specifically, an APDU command acceptable to an application after the personalization process has been completed, and the application selected
ARQC	Authorization Request Cryptogram
ATC	Application Transaction Counter
AUC	Application Usage Control
AuthC	Authorization Controls
BIN	Bank Identification Number
CA	Certification Authority

Acronym	Description
CAM	Card Authentication Method
Card	An IC payment card as defined by a payment system
Card Personalization	The personalization of application data within a card, using personalization commands
CBC	Cipher Block Chaining
CDA	Combined DDA/AC Generation
CDOL	Card Risk Management Data Object List
CID	Cryptogram Information Data
CLA	Class Byte
C-MAC	Command Message Authentication Code—MAC for secure communication during personalization command processing
CPLC	Card Production Life Cycle History File Identifiers
CPM	Common Personalization Method
CPS	Card Personalization Specification
CRT	Chinese Remainder Theorem
CSN	Chip Serial Number
CTAL	Cumulative Transaction Amount Limit
CTTA	Cumulative Total Transaction Amount
CVM	Cardholder Verification Method
CVN	Cryptogram Version Number

Acronym	Description
Data Preparation	The process of preparing and formatting data, ready for sending to a personalization device
dCVV	Dynamic Card Verification Value
DDA	Dynamic Data Authentication
DDOL	Dynamic Data Authentication Data Object List
DEK	Data Encryption Key
DEK/TK	Data Encryption Key / Transport key
DES	Data Encryption Standard
DGI	Data Group Identifier
ECB	Electronic Code Book
EMV	Europay, MasterCard, UICS
EMV CPS	EMV Card Personalization Specification, Personalization specification published by EMVCo
EMVCo LLC (EMVCo)	Payment system organization that manages, maintains, and strengthens EMV specification
ENC MDK	Main Data Encryption DEA Key
ENC UDK	Unique Data Encryption DEA Key
FCI	File Control Information
fDDA	FAST DDA
GP	Global Platform

Acronym	Description
GPO	Get Processing Options Command
hex	Hexadecimal
HSM	Hardware Security Module
IAC	Issuer Action Code
IAD	Issuer Application Data
IAP	Initiate Application Processing
Iauth	Issuer Authentication
IC	Integrated Circuit
ICC	Integrated Circuit Card
ICC Data	Input data record data element containing application DGIs
IEC	International Electrotechnical Commission
IIN	Issuer Identification Number
INS	Instruction Byte
IPK	Issuer Public Key
ISO	International Organization for Standardization
ISS	Issuer
K_{DEK}	Unique key used to generate session keys, this session key is used in the generation of C-MAC during command processing
KEK	Key Encryption Key

Acronym	Description
KEK/TK	Key Exchange Key/ Transport key - shared by data preparation device and personalization device
KEK _{ISS}	Key Exchange Key - shared by issuer and data preparation device
K _{ENC}	Unique card key used to generate encryption session key
K _{MAC}	Unique card key used to generate session key, this session key is used to encrypt DES key and other secret data (optional)
KMC	DES Master Key used as session key during personalization - used to generate derivation key, generate MAC key, and secret data protection keys(K _{ENC} , K _{DEK} , K _{MAC}), used to protect the personalization process
KMU	Issuer master key, held only by issuer, used to generate derivation key, generate MAC key and encryption key and secret data protection keys, re-personalization after card issuance is made possible using these keys
LCOL	Lower Consecutive Offline Limit
MAC	Message authentication code—card cryptogram in INITIALIZE UPDATE command
MAC MDK	Main message authentication code DEA key
MAC UDK	Unique message authentication code DEA key
MDK	Master Derivation Key
MSD	Magnetic Stripe Data format, refers to contactless payment using magnetic track 2 equivalent data
MSD Offset	MSD Offset

Acronym	Description
N/A	Not Applicable
PAN	Primary Account Number
Payment System	Payment System
UICS	China Financial IC Card Specification
PEK/TK	PIN Encryption Key/ Transport key
Personalization	Personalization, writing data into card to facilitate cardholder use of card
Personalization Device	Personalization Device
Personalization Command	Personalization Command
PIN	Personal Identification Number
PIX	Proprietary Identifier Extension
PK	Public Key
Pre-personalization	Pre-personalization, data initialization prior to personalization
PSE	Payment System Environment
qUICS	Quick Debit/Credit Application
Ref.	Reference
Req.	Requirement
RFU	Reserve for Future Use

Acronym	Description
RID	Registered Application Provider Identifier
RSA	Rivest, Shamir and Adleman Public Key Cryptographic Algorithm
SAD	Signed Static Application Data
SDA	Static Data Authentication
SFI	Short File Identifier
SKU	Personalization Session Key
SKU _{DEK}	Session key used for creating the C-MAC used in command processing - generated by K _{DEK}
SKU _{ENC}	Session key used for encryption - generated by K _{ENC}
SKU _{MAC}	Session key used to encrypt DES key and other secret data (optional) - generated by K _{MAC}
SUDK ENC	ENC generated unique data encryption session key
SUDK MAC	UDK generated MAC session key
TC	Transaction Certificate
TK	Transport key
TLV	Tag,Length,Value
UCOL	Upper Consecutive Offline Limit
UDK	Unique UICS card level key generated by MDK
Var.	Var.iable

4 Personalization Processing Overview

In order to assist members to easily configure card parameters upon card issuance, China UnionPay has produced the templates as guides for technology configuration. This document defines the minimum data set required for each template, and each data object requires a data value, the value is set either with the optimized recommendation or must be designated by issuer.

4.1 Card Personalization Data Processing

4.1.1 Processing Overview

Personalization process includes the following elements: Issuer data; cardholder related issuer data; certificates and keys required by application; magnetic stripe data; letters and special info on card; embossing of card data; command that controls personalization device [personalization device command flow (PDI)]; and hardware security module. Issuer and supplier both participate in the personalization process, determine the optimized organizational tasks and elements, follow product rules for using personalized products, but must execute these tasks in a secure fashion.

Personalization process consists of the following three functional segments:

- 1) Data preparation
- 2) Personalization device configuration and processing
- 3) IC card application processing

The steps and interfaces in these segments are described briefly here, while more detailed discussions are in following sections. More personalization process details are introduced in *EMV Card Personalization Specification*.

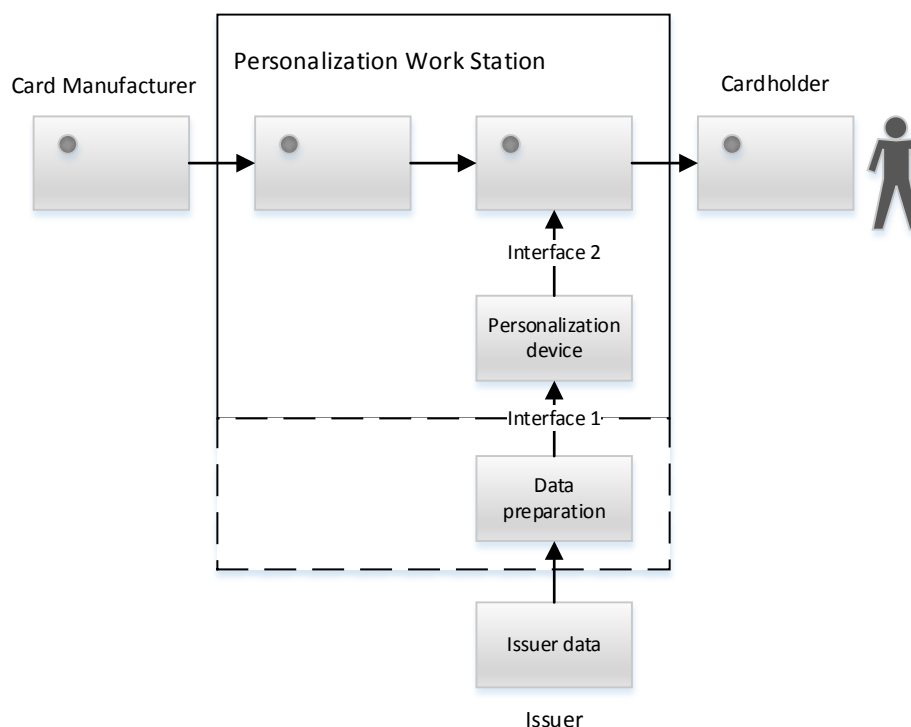


Figure 1 Personalization process flow overall map

- Data preparation:

Data preparation process refers to either retrieving relevant data from the issuer database or the issuer generating the data, then the data is processed and prepared for writing into the IC card during the personalization process.

Some data are the same across every card, and some others are different for each card. Some data may be always encrypted throughout the personalization process, data preparation may be a single process, or interact between different systems.

Most definitions regarding data preparation is based on specified applications. This document mainly focuses on the general IC card data preparation and processing. Part 3 describes a detailed UICS data preparation request.

- Interface 1: From data preparation to personalization device

The output of the data preparation process is a set of personalization data to be transferred to the personalization device. Data preparation system must protect the authenticity and integrity of personalization data (for example using MAC or hash signatures)

- Personalization device

Personalization device is a chip reader-writer that sends personalization data to the IC card. For the majority of IC card applications that use general methods for personalization, the device must connect to the IC card through the hardware security module (HSM), to facilitate data decryption and MAC verification (ex. EXTERNAL AUTHENTICATE command) when sending commands to the

application. Personalization device processing flow is described in *EMV Card Personalization Specification*, while Section 5.2 of this document describes the usage of keys in the personalization process.

Personalization device processing flow should be independently designed based on the application. Personalization process for all applications should be the same.

- Interface 2: From personalization device to IC card

Personalization device sends a series of personalization commands to the IC card, including id, command, and data. According to the result returned by IC card for each command, personalization device decides whether this command sent was successful or failed and processes accordingly. Personalization device must take appropriate measures when commands fail. Personalization command flow is listed in *EMV Card Personalization Specification*.

- IC card application

IC card application obtains personalization command from the personalization device, and stores the data in these commands in appropriate locations, so that these data could be used during application initiation.

EMV Card Personalization Specification describes the processing requirements for the IC card application prior to personalization. Pre-personalization IC card processing (pre-processing) is not within the scope of this document. Therefore, we assume that IC card application has all required keys for personalization prior to the personalization process.

Specific process could reference the descriptions in Appendix A.

4.1.2 Personalization Device

In the *EMV Card Personalization Specification* description, personalization process is designed to support application personalization on the card. It establishes a personalization basis framework that allows for application upgrading without changing personalization device. The entire personalization process can extend application to other same type applications.

This personalization basis framework is composed of the following segments:

- Security requirements between the personalization device and IC card. An overview is provided in Section 2.1.3.
- Command that sends personalization data to the IC card application. An overview is provided in Section 2.1.4.
- Recording format standards for sending to personalization device.

Appendix A describes the subsection of EMV card personalization specification for personalization of applications on qUICS cards, which is reserved for specific markets as described in this document.. Please contact your regional representative to determine whether Appendix A is a sufficient implementation of *EMV Card*

Personalization Specification for your product on the market. Suppliers that already support *EMV Card Personalization Specification* must support the cards that only use this subsection without changing existing systems.

4.1.3 Security Message

Prior to beginning personalization, personalization device must establish secure channel with IC card. The commands used to establish the secure channel include INITIALIZE UPDATE command and EXTERNAL AUTHENTICATE command, which are both described in *EMV Card Personalization Specification*.

Two sub-keys on the IC card are used in the process of establishing the secure channel. These are the encryption key (Kenc) and MAC key (Kmac). Process key SKUenc is generated by deriving Kenc, and through SKUenc generate and verify authentication cryptogram. Process key SKUmac is generated by deriving Kmac, and through SKUmac generate the MAC required by the EXTERNAL AUTHENTICATE command. These keys are all obtained by deriving master key KMC. IC card provides KMC ID and derivation data to the personalization device, and then the device obtains the sub-key through these data. *EMV Card Personalization Specification* contains description of KMC ID and subkey generation method. Once secure channel is established, personalization data could be sent to IC card applications.

4.1.4 STORE DATA

This command is used to send personalization data to card applications, and is described in detail in *EMV Card Personalization Specification*.

In order to reduce personalization time, data preparation process personalization device sends personalization data to the IC card application as data packets.

Data in some data packets must be sent by the personalization device to the card in cryptogram format. Personalization device and card can use mutually agreed upon keys for encryption processing. At this time, subkey Kdek on IC card is used to generate process key SKUdek. Subkey Kdek is the same as Kenc, Kmas, generated by deriving the master key KMC. IC card provides personalization device with KMC ID and derivation data, generated subkey (SKUdek) is used for encrypting data.

5 Data Preparation Requirements

Data preparation must create data used for application personalization . The Data created from the data preparation process must be securely transferred to the personalization device (unless the data is generated in the personalization device's cache). Secret data created from the data preparation process must be encrypted, to ensure the integrity of these data a MAC for data to be transferred to the personalization device should be generated.,.

Data preparation process includes the following steps:

- 1) Collecting personalization data from issuing channel
- 2) Grouping personalization data into data packets
- 3) Apply encryption and MAC protection for important personalization data
- 4) Create personalization command (this process is optional and can be completed by personalization device)
- 5) Create application required data
- 6) Create input file for personalization device

Personalization data is classified into 3 categories

- Issuer master key and relevant data
- Application key and certificate
- Application data

5.1 Issuer Master Key and Relevant Data

Personalization process generally requires the creation of issuer master key and relevant data. Master key is used to generate card or application key. Some data is used for managing personalization flow, and some data is stored into the card in the personalization process (refer to Part 5 for overview of keys used in application and personalization).

Other processing flows can also use one or more master keys provided for the personalization process. To ensure the master keys can be securely shared between processes, there must be a way to import and export master keys.

5.2 Application Key and Certificate

Symmetric DES key must be generated in the data preparation process. In most cases, these keys are obtained through derivation of issuer master key.

If UICS supports card authentication, issuer authentication, or issuer script processing, then application DES key must be derived from issuer master key using PAN and PAN sequence number.

If card supports offline data authentication (SDA, DDA, CDA or fDDA), the issuer must generate a pair of asymmetric issuer public/private keys, have a CA generate

certificate for issuer public key, then storing this issuer public key certificate in the card.

Also, cards that support dynamic data authentication (DDA, CDA, or fDDA) functionality must support RSA public key. Generating IC card RSA key pairs for every card, IC card public key must be signed by issuer private key, and generating corresponding IC card public key certificates. The generated IC card public key certificate and corresponding private key must be stored in the card.

It is recommended that if offline cryptogram PIN is supported, then IC card key pairs must support offline cryptogram PIN and DDA or CDA (currently, UICS specifications do not support offline cryptogram PIN). Please refer to Section 4 for overview of application key usage.

5.3 Application Data

Some application data may be usable for all IC cards from one issuer (ex. issuer id), or may be the same for specified cards in a certain batch (ex. UICS issuer action code), and some application data is unique for the IC card (ex. UICS application PAN).

6 Key Personalization Requirements

This section describes the entities and keys used in the personalization process. To clearly differentiate UICS application keys and common personalization keys, UICS application keys (personalized into card) are firstly described below.

6.1 UICS Application Key Overview

Table 6-1 describes the personalized UICS application key and optional post-personalization update key.

Table 6-1 UICS application key

Key Name	Key Sharing	Usage	Master Key	Card Key	Session Key
UICS online authentication key	Issuer and card	Master key is used to generate unique card key, used for online verification of card and issuer.	MDK	UDK	SUDK(Used in common cryptogram)
UICS message authentication key	Issuer and card	Master key is used to generate unique card key which is used to generate data authentication session key required for updating data post-issuance.	MAC MDK	MAC UDK	SUDK MAC
UICS data encryption key	Issuer and card	Master key is used to generate unique card key which is used to generate session key to encrypt post-issuance secret data updates (offline PIN).	ENC MDK	ENC UDK	SUDK ENC
ICC private key	Issuer and card	Generated by issuer and securely stored on card. In the offline data authentication (DDA) processing, this private key is used to sign dynamic data. After completing personalization, issuer generally does not hold this key.			

Key Name	Key Sharing	Usage	Master Key	Card Key	Session Key
ICC PIN encryption private key	Issuer and card	Issuer generates and securely stores in card, if ICC private key is not used, then use for decryption offline cryptogram PIN. Generally the issuer does not retain this key after personalization			

6.2 UICS Personalization Key Overview

KMC is used to generate the static unique sub-key in the card, this key is used to generate session key for IC card application communications. Table 5-2 summarizes UICS personalization related keys.

Table 6-2 UICS personalization keys

Key Name	Key Sharing	Usage	Master Key	Card Key	Session Key
Issuer master key	Issuer, IC card manufacturer, and personalization device	IC card manufacturer uses this KMC to generate card level keys (KENC, KMAC, KDEK), and writes them into card.	KMC		
		Used to create a session key which is used to create cryptogram and use CBC mode encryption of secret data.		K _{ENC}	SKU _{ENC}
		Used to create a session key which is used to create C-MAC in command processing.		K _{MAC}	SKU _{MAC}
		Used to create a session key which is used to encrypt symmetric keys or flexibly encrypt other secret data in ECB mode		K _{DEK}	SKU _{DEK}

Key Name	Key Sharing	Usage	Master Key	Card Key	Session Key
Issuer key exchange key	Issuer and data preparation device	Protect offline PIN and other secret data between issuer and data preparation device.	KEK _{ISS}		
Data encryption key/ transport key	Data preparation device and personalization device	Protect offline PIN and other secret data between data preparation device and personalization device. The following special type data transport keys may be used: --PEK/TK: PIN encryption key, used to protect PIN data; --KEK/TK: key exchange key, used to protect symmetric keys.	DEK/TK		
MAC key (verification code key)	In personalization data file, provided by data preparation device to personalization device	Used to ensure integrity of application data provided to personalization device in personalization data file.	MAC Key	N/A	N/A

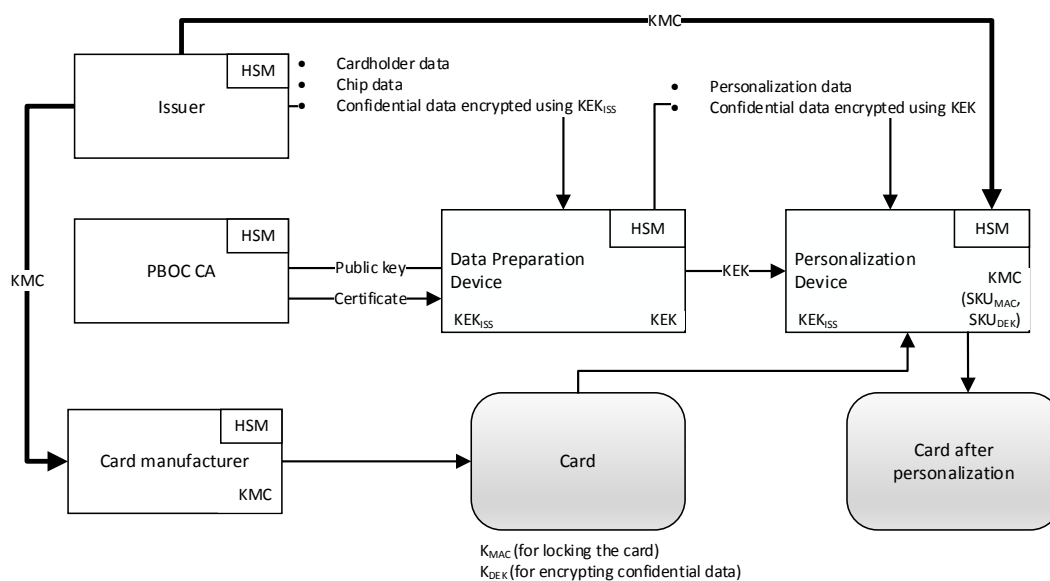


Figure 2 UICS personalization key usage

For specific usage methods of keys in personalization please see Appendix A.

7 Personalization General Requirements

7.1 Personalization General Requirements

Using general personalization method (see Appendix B) to personalize UICS applications must meet the following requirements:

- 1) Unless otherwise specified in this Specification, personalization processing must meet all requirements in *EMV Card Personalization Specification*.
- 2) Card application must meet all requirements in this application specification (ex. UICS, qUICS).
- 3) Mandatory data in application specifications (ex. UICS, qUICS) must be personalized in card application.
- 4) Card application can ignore the encryption bit in the P1 parameter of the STORE DATA command, and can directly decrypt the encrypted DGI. Please see description in 5.3.3.

7.2 Data Packet Requirements

In the personalization flow, data packets play an important role. Personalization data is logically or physically packaged and recognized through data grouping identifiers (DGI). Personalization device resolves input data and create a corresponding STORE DATA command for each data packet in input records. All data in a packet (within one DGI) is stored in card application through STORE DATA command. After personalization device receives the data packet, IC card application determines how to process this data packet through DGI. Data packets are defined in Sections 7, 8, 9, 10, and 11.

Data grouping identifier (DGI) is represented through 2 byte hexadecimal. For specific format please see 5.3.2 STORE DATA command.

- 1) All data elements that need to be written into the IC card must be placed in a certain data packet (DGI).
- 2) Key storage data packet could only store keys.
- 3) Data grouping identifier byte 1 from '01' to '1E' represents the SFI of data stored. For these data grouping identifiers, byte 2 represents SFI record's record number. The data file ids that doesn't begin with '01' to '1E' are used to represent internal data.
- 4) Byte 1 from '01' to '0A' data packets shall include a record template, with a tag '70', its length is the length of record data.
- 5) Within a data packet, data length shall not exceed 237 bytes.
- 6) The two DGIs '0D01' and '0E01' are used for personalization of internal application data. They are stored in specific region of the card application that could not be accessed using READ RECORD or UPDATE RECORD commands.

- 7) Data grouping identifier length is generally represented by 1 byte binary. Tag length is generally represented using 1 byte, unless length exceeds 127 bytes, in which case a byte '81' would be added before the length byte. For example, a tag value that includes 128 bytes is '8180'.
- 8) Data packets '9F6B' to '9F6F' is reserved for payment organization, which generally uses for personalizing some proprietary data rates, such as card manufacturing data, etc.
- 9) Data packets '7FF0' to '7FFE' are reserved for personalization processes unrelated to application.
- 10) Data packets '8F01', '7F01', and '00CF' are reserved for derivation key that creates secure channel and key related data.
- 11) Data packet '0062' is reserved for file creation.
- 12) Number of data packets is kept at a minimum so as to improve personalization efficiency.
- 13) Key verification value is generally used to verify whether key installation is correct, generally the value is obtained by using the installed key to calculate 8 bytes of '00' using ECB 3DES key, the leftmost 3 bytes of the calculation result is the key verification value.

7.3 Data Packet List

Personalization data packets used by application are partially defined by [EMV personalization specification], for example, PSE, key, and PIN related data, select command response, GPO command response (no electronic cash transaction), while others are defined in this Specification. This Specification defines DGI packets not specified in [EMV personalization specification].

7.4 Personalization Command Requirements

Personalization process uses the following commands:

- 1) SELECT
- 2) INITIALIZE UPDATE
- 3) EXTERNAL AUTHENTICATE
- 4) STORE DATA

For specific command format and content, please see Appendix A.

7.5 Common AID Personalization Requirements

If card payment application (usually debit application) supports Common AID, then it shall create two instances in personalization: The first instance uses UnionPay debit card application, and personalize all debit application packets under this AID (refer to Chapter 8, Chapter 9, except PSE/PPSE); The second instance uses Common AID, and personalize below packets under this AID:

DGI Packet	Content
9102	In Contact interface, application select FCI information
9103	In Contactless interface, application select FCI information

Besides, if card payment application (usually debit application) also supports Common AID, then application directory entry (PSE and PPSE) shall support both standard AID and common AID, as follows:

PSE Directory File Record Packet '010n'	PSE directory file shall personalize two record: the first one is UnionPay debit card AID, the other one is Common AID, details refer to 8.1.2
Select PPSE Application Reponse Data '9102'	PPSE FCI information shall personalize both standard debit card application entry and Common AID application entry, details refer to 9.1.1

Note: this chapter only provides the minimum Common AID Personalization requirements; application provider can add other packets as needed.

8 Debit/Credit Application (Including Low-value Payment) DGI Packets

The personalization DGI packet within this chapter can be either standard UICS Debit/Credit data packet or low-value payment UICS data packet.

8.1 PSE Data Packets

8.1.1 DGI 9102: Select PSE Application Response Data (FCI)

Tag		Data Element	Length	
A5		FCI custom template	Var.	Mandatory
	88	Directory elementary file SFI	1	Mandatory
	5F2D	Preferred language	2–8	Recommended
	9F11	Issuer code table index	1	Optional
	BF0C	Issuer discretionary data	Max 222	Optional

8.1.2 DGI:010n: Directory File Record Packets

Tag		Data Element	Length	
61		FCI custom template	Var.	Mandatory
	4F	Application identifier (AID)	5–16	Mandatory
	50	Application tag	1 – 16	Recommended
	9F12	Application priority name	1 –16	Optional
	87	Application priority indicator	1	Recommended
	73	Directory self-defined template	Max 252	Optional

8.2 Standard Record Packets

The certificate and public key related data in this section's standard record packet are based on RSA algorithm system.

All the record packets and its related layout are not mandatory within this specification, issuer or personalization organizations can adjust or format record packet and AFL according to their requirements, but they need to make sure AFL can correctly refer to record data packets defined by specifications.

8.2.1 DGI 0101: Cardholder basic information data (1)

Tag	Data Element	Length	Note
57	Magnetic stripe 2 equivalent data	Max 19	Mandatory
9F1F	Magnetic stripe 1 self-defined data	Var.	Optional
9F61	Cardholder personal identification number	1-40	Mandatory
9F62	Cardholder personal identification type	1	Mandatory
5F20	Cardholder name	2-26	Mandatory

8.2.2 DGI 0102: Cardholder basic information data (2)

Tag	Data Element	Length	Note
9F61	Cardholder personal identification number	1-40	Mandatory
9F62	Cardholder personal identification type	1	Mandatory
5F20	Cardholder name	2-26	Mandatory

Note: If issuer requires cardholder basic info to be successfully updated through online script method, then cardholder basic info can be personalized into DGI 0102.

8.2.3 DGI 0201: Issuer Public Key Certificate

Tag	Data Element	Length
90	Issuer public key (IPK) certificate	Var.

8.2.4 DGI 0202: Issuer Public Key Data

Tag	Data Element	Length
8F	CA public key index	1
9F32	Issuer Public Key Exponent	1or 3
92	Issuer public key remainder	Var.

8.2.5 DGI 0203: Static Application Data Signature

Tag	Data Element	Length
93	Static application data signature	Var.

8.2.6 DGI 0204: IC Card Public Key Certificate

Tag	Data Element	Length
9F46	IC card public key certificate	Var.

8.2.7 DGI 0205: IC Card Public Key Data

Tag	Data Element	Length
9F47	IC card public key index	1 or 3
9F48	IC card public key remainder	Var.
9F49	Dynamic data authentication object list DDOL	Var.

8.2.8 DGI 02nn: IC Card Public Key Certificate (Multiple)

Tag	Data Element	Length
9F46	IC card public key certificate	Var.

Note: If IC Card supports both low-value payment and contactless application, then the static application data that take part in signature may have more than one values, thus more than one public key certificate are needed.

8.2.9 DGI 02nn: Static Application Data Signature (Multiple)

Tag	Data Element	Length
93	Static application data signature	Var.

Note: If IC Card supports low-value payment, then the static application data that take part in signature may have more than one values, thus more than one public key certificate are needed.

8.2.10 DGI 0301: Static Signature Authentication Data

Tag	Data Element	Length
5F24	Application Expiration Date	3
5F25	Application effective date	3
5A	Primary account number (PAN)	Var. (Max 10)
9F07	Application usage control	2
8E	Cardholder verification method (CVM) list	Var.
9F0D	Issuer action code (IAC) - Default	5
9F0E	Issuer action code (IAC) - Reject	5

Tag	Data Element	Length
9F0F	Issuer action code (IAC) - Online	5
5F28	Issuer country code	2
9F08	Application version number (Optional)	2
9F4A	Static data authentication tag list	Var.
8C	Card risk management data object list 1 (CDOL1)	Var.
8D	Card risk management data object list 2 (CDOL2)	Var.
5F34	Application main account serial number (Optional)	1

Notes: In static signature authentication data, if certain data supports script update, or has multiple values in different path, then these data are either excluded from static signature authentication data, or multiple static signature authentication data need to be personalized. Besides, if card supports qUICS path and application primary account serial number '5F34' is returned in GPO command response data, then '5F34' should not be included in this static signature authentication data.

8.2.11 DGI 0302: Card Administrative Data

Tag	Data Element	Length
5F30	Service code	2
9F42	Application currency code	2
9F63	Product Label Information	16
9F08	Application version number (Optional)	2

Notes: If '9F08' is part of static signature authentication data, then '9F08' is not returned in this packet.

8.2.12 DGI 03nn: Static Signature Authentication Data (Multiple)

Tag	Data Element	Length
5F24	Application Expiration Date	3
5F25	Application effective date	3
5A	Primary account number (PAN)	Var. (Max 10)
9F07	Application usage control	2
8E	Cardholder verification method (CVM) list	Var.
9F0D	Issuer action code (IAC) - Default	5
9F0E	Issuer action code (IAC) - Reject	5
9F0F	Issuer action code (IAC) - Online	5
5F28	Issuer country code	2
9F08	Application version number (Optional)	2
9F4A	Static data authentication tag list	Var.
8C	Card risk management data object list 1 (CDOL1)	Var.
8D	Card risk management data object list 2 (CDOL2)	Var.

Tag	Data Element	Length
5F34	Application main account serial number (Optional)	1

Notes: In static signature authentication data, if certain data supports script update, or has multiple values in different path, then these data are either excluded from static signature authentication data, or multiple static signature authentication data need to be personalized. Besides, if card supports qUICS path and application primary account serial number '5F34' is returned in GPO command response data, then '5F34' should not be included in this static signature authentication data.

8.2.13 DGI 0401: Terminal Risk Management

Tag	Data Element	Length
9F14	Consecutive offline transactions lower limit	1
9F23	Consecutive offline transactions upper limit	1

8.2.14 DGI 0402: Electronic cash authorization code (low-value payment)

Tag	Data Element	Length
9F74	Electronic cash authorization code	6

Note1: If the transaction path is low-value payment, then the corresponding AFL design should contain this DGI packets and the DGI should be ensured to be AFL's last record.

8.3 Internal Data Packets

General internal packet sharing status is as shown in the table below:

8.3.1 DGI 0D01: Card Internal Risk Management Data

Tag	Data Element	Length	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
9F58	Consecutive offline transactions lower limit	1	√		
9F59	Consecutive offline transactions upper limit	1	√		
9F53	Consecutive offline transaction limit, international (currency)	1	√		
9F72	Consecutive offline transaction limit, international (country)	1	√		
9F54	Cumulative offline transaction amount limit	6	√		
9F75	Cumulative offline transaction amount limit (dual currency)	6	√		
9F73	Currency conversion factor	4	√		
9F5C	Cumulative offline transaction amount upper limit	6	√		
9F77	Electronic cash balance upper limit	6		√	√
9F78	Electronic cash single transaction limit	6		√	√

Tag	Data Element	Length	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
9F79	Electronic cash balance	6		√	√
9F6D	Electronic cash reset threshold	6		√	√

If financial UICS application must support dual currency electronic cash applications, then 0D01 shall include the tags as follows:

Tag	Data Element	Length	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
DF71	Secondary currency electronic cash application currency code	2		√	√
DF76	Secondary currency electronic cash reset threshold	6		√	√
DF77	Secondary currency electronic cash balance upper limit	6		√	√
DF78	Secondary currency electronic cash single transaction limit	6		√	√
DF79	Secondary currency electronic cash balance	6		√	√

Notes: this 0D01 packet can be installed by many commands. In future version of specification, this packet will be moved to 3000 or 3001 packet, which is consistent with EMVCo CPS specification.

8.3.2 DGI 0E01: Card Internal Risk Management Data

Tag	Data Element	Length	Condition	Standard Debit/Credit	Low-value payments Based On Debit/Credit
9F51	Application currency code	2	Mandatory	√	√
9F52	Application default action (ADA)	2	Mandatory	√	√
9F56	Issuer authentication indicator bit	1	Mandatory	√	√
9F57	Issuer country code	2	Mandatory	√	√
9F76	Secondary application currency code	2	Optional	√	√
9F36	Application transaction counter (ATC)	2	Present when 3000 packet doesn't exist. Notes: In the future version of this specification, this data will be stored in 3000 packet.	√	√
9F4F	Transaction log format	Var.	Present when 3000 packet doesn't exist. Notes: In the future version of this specification, this data will be stored in 3000 packet.		√

Tag	Data Element	Length	Condition	Standard Debit/Credit	Low-value payments Based On Debit/Credit
DF4F	Transaction log format	Var.	Present when 3000 packet doesn't exist Notes: In the future version of this specification, this data will be stored in 3000 packet.		

Notes: this 0D01 packet can be installed by many commands. In future version of specification, this packet will be moved to 3000 or 3001 packet, which is consistent with EMVCo CPS specification.

8.3.3 DGI 3000: ATC and Log Format (Contact)

Tag	Data Element	Length	Condition
9F36	Application transaction counter (ATC)	2	Mandatory
9F4F	Transaction log format	Var.	Mandatory
DF4F	Circular log format	Var.	Optional

Note: 3000 packet data elements could also be stored in 0D01 or 0E01 DGI packets.

8.4 Key Data Packets

General internal packet sharing status is as shown in the table below:

Tag	Data Element	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
8000	3DES symmetric key	√	√	√

Tag	Data Element	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
9000	3DES symmetric key verification value	√	√	√
8010	Offline PIN	√	√	
9010	Offline PIN related data	√	√	
8101/8103	RSA ND format key	√	√	√
8201~8205	RSA CRT format key	√	√	√

8.4.1 DGI 8000:3DES Symmetric Key

Tag	Data Element	Length	Encrypted
-	Application cryptogram DEA key (UDK)	16	Yes
-	Message authentication code DEA key (MAC UDK)	16	Yes
-	Data encryption DEA key (ENC UDK)	16	Yes

8.4.2 DGI 9000:3DES Symmetric Key Verification Value (Recommended)

Tag	Data Element	Length	Encrypted
-	Application cryptogram DEA key verification value	3	No
-	Message authentication code DEA key verification value	3	No

Tag	Data Element	Length	Encrypted
-	Data encryption DEA key verification value	3	No

8.4.3 DGI 8010: Offline PIN

Tag	Data Element	Length	Encrypted
-	Offline PIN	8	Yes

Note: Please refer to UICS specification requirements for specific offline PIN pre-encryption format, encryption key can use SKU_{DEK}

8.4.4 DGI 9010: Offline PIN Related Data

Tag	Data Element	Length	Encrypted
-	PIN retry counter	1	No
-	PIN retry limit	1	No

8.4.5 DGI 8101/8103:RSA CLASSIC Private Key

DGI	Data Element	Length	Encrypted
8101	Private key D component	Var.	Yes
8103	Private key N component	Var.	Yes

8.4.6 DGI 8201-8205:RSA CRT Private Key

DGI	Data Element	Length	Encrypted
8201	Private key Qinv component ($q-1 \bmod p$)	Var.	Yes
8202	Private key Dq component	Var.	Yes

DGI	Data Element	Length	Encrypted
8203	Private key Dp component	Var.	Yes
8204	Private key Q component	Var.	Yes
8205	Private key P component	Var.	Yes

Equivalent mode:

DGI	Data Element	Length	Encrypted
8201	Private key (P-1 mod Q) component	Var.	Yes
8202	Private key Dp component	Var.	Yes
8203	Private key Dq component	Var.	Yes
8204	Private key P component	Var.	Yes
8205	Private key Q component	Var.	Yes

Note1: 8201~8205 definition follow the *EMV Card Personalization Specification*, no mode besides these two would be permitted;

Note2: Either ND mode grouping or CRT mode grouping as defined in 6.37 could be selected.

8.5 Command Response Packets

8.5.1 DGI 9102: Select response data FCI

Tag	Data Element	Length	Condition
A5	FCI custom template	Var..	Mandatory
50	Application tag	1 – 16	Mandatory

Tag		Data Element	Length	Condition
	87	Application priority indicator	1	Recommended
	9F38	PDOL	Var..	Recommended
	5F2D	Language preference	2 – 8	Optional
	9F11	Issuer code table index	1	Recommended
	9F12	Application priority name	1 -- 16	Optional
	BF0C	FCI Issuer discretionary data	Var.	Optional
	9F4D	Log entry	2	Optional
	DF4D	Circular log entry	2	Optional

8.5.2 DGI 9200: Issuer Application Data

Tag	Data Element	Length
9F10	Issuer application data	Var.

8.5.3 DGI 9104: GPO Response Data (UICS Debit/Credit Path)

Tag	Data Element	Length	Encrypted
82	AIP	2	No
94	AFL	Var.	No

Note: Data record packets and AFL definition could be assigned value according to actual circumstances, but AFL design should not ensure there are no duplicate data records.

8.5.4 DGI 9203: GPO Response Data (Low-value Payment Path)

Tag	Data Element	Length	Encrypted
82	AIP	2	No
94	AFL	Var.	No

Note: According to the definition of data record DGI packets in this Specification, electronic cash AFL value is "08010100 10040701 18010200", but in actual personalization operation, data record packets and AFL definition could be assigned value according to actual circumstances.

9 qUICS Path (Include Extended Application) DGI Packets

9.1 PPSE Data Packets

9.1.1 DGI 9102: Select PPSE Application Response Data FCI

Tag		Data Element		Length	
A5		FCI custom template		Var.	Mandatory
	BF0C		Issuer discretionary data	Max 222	Mandatory
		61	Application template	Var.	Mandatory
		4F	Application identifier (AID)	5–16	Mandatory
		50	Application tag	1 – 16	Recommended
		87	Application priority indicator	1	Recommended
		61	Application template	Var.	Mandatory (if there are multiple AIDs)
		...			

9.2 Standard Record Packets

The certificate and public key related data in this section's standard record packet are based on RSA algorithm system. If card supports the certificate system of SM2 algorithm, the public key related data of SM2 algorithm should be created during personalization.

All the record packets and its related layout are not mandatory within this specification, issuer or personalization organizations can adjust or format record packet and AFL according to their requirements, but they need to make sure AFL can correctly refer to record data packets defined by specification.

9.2.1 DGI 0201: Issuer Public Key Certificate

Tag	Data Element	Length
90	Issuer public key (IPK) certificate	Var.

9.2.2 DGI 0202: Issuer Public Key Data

Tag	Data Element	Length
8F	CA public key index	1
9F32	Issuer Public Key Exponent	1or 3
92	Issuer public key remainder	Var.
9F46	IC card public key certificate	Var.
9F47	IC card public key index	1 or 3
9F48	IC card public key remainder	Var.
9F49	Dynamic data authentication object list DDOL	Var.

Note: If qUICS and standard Debit/Credit use the same IC card public key certificate, then qUICS and standard Debit/Credit share this 0202 packets, or else, qUICS uses 02nn packets.

9.2.3 DGI 02nn: Issuer Public Key Data (Multiple)

Tag	Data Element	Length
8F	CA public key index	1
9F32	Issuer Public Key Exponent	1or 3
92	Issuer public key remainder	Var.

Tag	Data Element	Length
9F46	IC card public key certificate	Var.
9F47	IC card public key index	1 or 3
9F48	IC card public key remainder	Var.
9F49	Dynamic data authentication object list DDOL	Var.

Note: If qUICS and standard Debit/Credit use different IC card public key certificate, then qUICS uses 02nn packets.

9.2.4 DGI 0301: Static Signature Authentication Data

Tag	Data Element	Length
5F24	Application Expiration Date	3
5F25	Application effective date	3
5A	Primary account number (PAN)	Var. (Max 10)
9F07	Application usage control	2
8E	Cardholder verification method (CVM) list	Var.
9F0D	Issuer action code (IAC) - Default	5
9F0E	Issuer action code (IAC) - Reject	5
9F0F	Issuer action code (IAC) - Online	5
5F28	Issuer country code	2

Tag	Data Element	Length
9F08	Application version number (Optional)	2
9F4A	Static data authentication tag list	Var.
8C	Card risk management data object list 1 (CDOL1)	Var.
8D	Card risk management data object list 2 (CDOL2)	Var.
5F34	Application main account serial number (Optional)	1

Note: If qUICS and standard Debit/Credit use the same static signature authentication data, then qUICS AFL should contain this packet, or else, qUICS uses 03nn packets.

9.2.5 DGI 03nn: Static Signature Authentication Data (Multiple)

Tag	Data Element	Length
5F24	Application Expiration Date	3
5F25	Application effective date	3
5A	Primary account number (PAN)	Var. (Max 10)
9F07	Application usage control	2
8E	Cardholder verification method (CVM) list	Var.
9F0D	Issuer action code (IAC) - Default	5

Tag	Data Element	Length
9F0E	Issuer action code (IAC) - Reject	5
9F0F	Issuer action code (IAC) - Online	5
5F28	Issuer country code	2
9F08	Application version number (Optional)	2
9F4A	Static data authentication tag list	Var.
8C	Card risk management data object list 1 (CDOL1)	Var.
8D	Card risk management data object list 2 (CDOL2)	Var.
5F34	Application main account serial number (Optional)	1

Note: If qUICS and standard Debit/Credit use different static signature authentication data, then qUICS uses 03nn packets as static signature authentication data.

9.2.6 DGI 03nn: Card Administrative Data

Tag	Data Element	Length
9F63	Product identification information	16
9F08	Application version number (Optional)	2
9F74	Electronic cash authorization code	6

Notes: If '9F08' is part of static signature authentication data, then '9F08' is not returned in this packet.

9.2.7 DGI 03nn: IC Card long Signature Record Return

Tag	Data Element	Length
9F4B	Dynamic Signature Data	Var
9F5D	Offline Purchase Balance	6

Notes1: This DGI can be configured by card manufacture via card personalization parameters.

Notes2: This DGI is generate by IC Card, and only presents when RSA ICC key is longer than 1024, don't need personalization. If dynamic signature data and offline purchase balance exceed one record length, then card should be separated into two records and store these data, qUICS AFL design should include this DGI packet.

9.2.8 DGI 03nn: Card Authentication Related Data

Tag	Data Element	Length
9F69	Card Authentication Related Data	Var

Notes1: If current transaction uses 01 version fDDA algorithm to sign a signature, then qUICS AFL should be included to make sure this DGI is the last record of AFL.

9.3 Internal Data Packets

9.3.1 DGI 0D01: Card Internal Risk Management Data

Tag	Data Element	Length	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
9F58	Consecutive offline transactions lower limit	1	√		
9F59	Consecutive offline transactions upper limit	1	√		

Tag	Data Element	Length	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
9F53	Consecutive offline transaction limit, international (currency)	1	√		
9F72	Consecutive offline transaction limit, international (country)	1	√		
9F54	Cumulative offline transaction amount limit	6	√		
9F75	Cumulative offline transaction amount limit (dual currency)	6	√		
9F73	Currency conversion factor	4	√		
9F5C	Cumulative offline transaction amount upper limit	6	√		
9F77	Electronic cash balance upper limit	6		√	√
9F78	Electronic cash single transaction limit	6		√	√
9F79	Electronic cash balance	6		√	√
9F5D	Available offline purchasing amount	6			√
9F6D	Electronic cash reset threshold	6		√	√

Tag	Data Element	Length	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
9F6B	Card CVM limit	6			√
9F68	Card additional processing	4			√
9F6C	Card transaction attribute	2			√
9F5D	Available offline purchasing amount	6			√

If financial UICS application must support dual currency electronic cash applications, then 0D01 shall include the tags as follows:

Tag	Data Element	Length	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
DF71	Secondary currency electronic cash application currency code	2		√	√
DF72	Secondary currency card CVM limit	6			√
DF76	Secondary currency electronic cash reset threshold	6		√	√
DF77	Secondary currency electronic cash balance upper limit	6		√	√
DF78	Secondary currency electronic cash single transaction limit	6		√	√

Tag	Data Element	Length	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
DF79	Secondary currency electronic cash balance	6		√	√

If extended low-value payment applications are supported, the 0D01 shall include the tags as follows:

Tag	Data Element	Length	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
DF62	Electronic cash deduction by segment limit	6			√
DF63	Electronic cash deducted by segment	6			√
DF7A	Second currency e-cash segment fee cost deduction limit	6			√
DF7B	Second currency e-cash segment fee cost deducted amount	6			√

9.3.2 DGI 0E01: Card Internal Risk Management Data

Tag	Data Element	Length	Condition	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
9F51	Application currency code	2	Mandatory	√	√	

Tag	Data Element	Length	Condition	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
9F52	Application default action (ADA)	2	Mandatory	√	√	
9F56	Issuer authentication indicator bit	1	Mandatory	√	√	
9F57	Issuer country code	2	Mandatory	√	√	
9F76	Secondary application currency code	2	Optional	√	√	
57	Magnetic stripe 2 equivalent data	Max 19	If present, then return in qUICS application GPO command response			√
5F20(9F0B)	Cardholder name	Var.	If present, then return in qUICS application GPO command response			√
5F34	Application primary account serial number	1	If present, then return in qUICS application GPO command online response or in number 17 cryptogram GPO response			√

Tag	Data Element	Length	Condition	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
DF63	Electronic cash deducted by segment	16	If present, then return in qUICS application GPO command online response or in number 17 cryptogram GPO response			√
9F36	Application transaction counter (ATC)	2	Present when 3000 packet doesn't exist. Notes: In the future version of this specification, this data will be stored in 3000 packet.	√	√	√
9F4F	Transaction log format	Var.	Present when 3000 packet doesn't exist. Notes: In the future version of this specification, this data will be stored in 3000 packet.	√	√	√

Tag	Data Element	Length	Condition	Standard Debit/Credit	Low-value payments Based On Debit/Credit	Quick Debit/Credit
DF4F	Transaction log format	Var.	Present when 3000 packet doesn't exist. Notes: In the future version of this specification, this data will be stored in 3000 packet.		√	√

9.3.3 DGI A001: Extended Application File Allocation Table (If Support Extended Application)

DGI	Data Element	Length	Condition
A001		Var.	Recommended
	Extended application file 1	7	Optional
	Extended application file 2	7	Optional
	...		

Extended application file's configuration item is organized as a unit by every 7 Bytes; each unit is an extended application file configuration, shown as below:

File Type	Data Unit	Length	Condition	Encryption
Linear variable length file	Extended application file SFI	1	Mandatory	NO
	File Type: 01 variable length record file 02 circulated record file	1	Mandatory	NO
	RFU:00	1	Mandatory	NO
	RFU:00	1	Mandatory	NO
	Record max length	1	Mandatory	NO
	File Size: for variable length files, 2 Byte represents default length; 00 00 represents no limit; for circulated files, have 1 Byte	2	Mandatory	NO

	record item number and 1 Byte record size			
--	---	--	--	--

Notes: all configuration above are suggested format, implementer can do in any format.

9.4 Key Data Packets

9.4.1 DGI 8000: 3DES Symmetric Key

Tag	Data Element	Length	Encrypted
-	Application cryptogram DEA key (UDK)	16	Yes
-	Message authentication code DEA key (MAC UDK)	16	Yes
-	Data encryption DEA key (ENC UDK)	16	Yes

9.4.2 DGI 9000: 3DES Symmetric Key Verification Value (Recommended)

Tag	Data Element	Length	Encrypted
-	Application cryptogram DEA key verification value	3	No
-	Message authentication code DEA key verification value	3	No
-	Data encryption DEA key verification value	3	No

9.4.3 DGI 8010: Offline PIN

Tag	Data Element	Length	Encrypted
-	Offline PIN	8	Yes

Note: Please refer to UICS specification requirements for specific offline PIN pre-encryption format, encryption key can use SKU_{DEK}

9.4.4 DGI 9010: Offline PIN Related Data

Tag	Data Element	Length	Encrypted
-	PIN retry counter	1	No
-	PIN retry limit	1	No

9.4.5 DGI 8020: Extended Application Access Key 3DES (If Support Extended Application)

Tag	Data Element	Length	Encrypted
-	Extended application access protection key for extended application file 1	16	Yes
-	Extended application access protection key for extended application file 2	16	Yes
-	Extended application access protection key for extended application file 3	16	Yes
	...		

The number and ordering of application access protection keys is determined based on the configuration data number and ordering in the A001 data packet. Ex: A001 data value is:

16 01 00 00 FF 00 00

19 01 00 00 FF 00 00

1E 02 00 00 80 0A 80

8020 data field length is $16 \times 3 = 48$ bytes, each 16 byte cryptogram is an access key, corresponding to 0x15 file access key respectively, 0x19 file access key, and 0x1E file access key.

9.4.6 DGI 9020: Extended Application Access Key 3DES Verification Value (If Support Extended Application)

Tag	Data Element	Length	Encrypted
-	Access key KCV for extended application file 1	3	No
-	Access key KCV for extended application file 2	3	No
-	Access key KCV for extended application file 3	3	No
	...		

9.4.7 DGI 8101/8103:RSA CLASSIC Private Key

DGI	Data Element	Length	Encrypted
8101	Private key D component	Var.	Yes
8103	Private key N component	Var.	Yes

9.4.8 DGI 8201-8205:RSA CRT Private Key

DGI	Data Element	Length	Encrypted
8201	Private key Qinv component ($q^{-1} \bmod p$)	Var.	Yes
8202	Private key Dq component	Var.	Yes
8203	Private key Dp component	Var.	Yes
8204	Private key Q component	Var.	Yes

DGI	Data Element	Length	Encrypted
8205	Private key P component	Var.	Yes

Equivalent mode:

DGI	Data Element	Length	Encrypted
8201	Private key (P-1 mod Q) component	Var.	Yes
8202	Private key Dp component	Var.	Yes
8203	Private key Dq component	Var.	Yes
8204	Private key P component	Var.	Yes
8205	Private key Q component	Var.	Yes

Note1: 8201~8205 definition follow the *EMV Card Personalization Specification*, no mode besides these two would be permitted;

Note2: Either ND mode grouping or CRT mode grouping as defined in 6.37 could be selected

9.4.9 DGI 8401 ~ DGI 8404: CA Public Key (Enhanced micropayment two-way authentication)

CA public key data is stored in the 8101~8104 DGI packets in order. 4 CA public keys are personalized mostly, and each packet stores one CA public key data. CA public key data storage format are as in table below.

Tag	Data Element	Length
--	CA key index	1
--	Public key index length	1
--	Public key index	1 or 3

Tag	Data Element	Length
--	Public key length	1
--	Public key	Var.

Note: Some enhanced micropayment products that do not follow requirement to use 8401~8404 packets for CA public key data could continue to be used, but prior to new product testing or publishing they should be re-designed according to this requirement.

9.5 Command Response Packets

9.5.1 DGI 9102: Select Application Response Data FCI (Contact or Contactless)

If same 'Select' command under contact interface or contactless interface is used, then general 9102 data packets can be used; If return with different 'Select' command under contact interface or contactless interface, then can use different data packets. If contact then use 9102 packets, if contactless then use 9103 packets.

Tag		Data Element	Length	Condition
A5		FCI custom template	Var..	Mandatory
	50	Application tag	1 – 16	Mandatory
	87	Application priority indicator	1	Recommended
	9F38	PDOL	Var..	Recommended
	5F2D	Language preference	2 – 8	Optional
	9F11	Issuer code table index	1	Recommended
	9F12	Application priority name	1 -- 16	Optional

Tag			Data Element	Length	Condition
	BF0C		FCI Issuer discretionary data	Var.	Optional
		9F4D	Log entry	2	Optional
		DF4D	Circular log entry	2	Optional

9.5.2 DGI 9103: Select Application Response Data FCI (contactless)

Tag		Data Element	Length	Condition	
A5		FCI customized template	Var..	Mandatory	
	50		Application tag	1 – 16	Mandatory
	87		Application priority indicator	1	Recommended
	9F38		PDOL	Var..	Recommended
	5F2D		Language preference	2 – 8	Optional
	9F11		Issuer code table index	1	Recommended
	9F12		Application priority name	1 -- 16	Optional
	BF0C		FCI Issuer discretionary data	Var.	Optional
		9F4D	Log entry	2	Optional
	DF4D	Circular log entry	2	Optional	

Tag			Data Element	Length	Condition
		DF61	Contactless added value application indicator	Var.	Mandatory, If extended application is supported

9.5.3 DGI 9200: Issuer Application Data

Tag	Data Element	Length
9F10	Issuer application data	Var.

Notes: In 9200 packet, algorithm identifier is 3DES.

9.5.4 DGI 9207: GPO Response Data (qUICS)

Tag	Data Element	Length	Encrypted
82	AIP	2	No
94	AFL	Var.	No
9F10	Issuer application data	Var.	

Note1: Data record packet layout and AFL value should follow actual situation, AFL design should make sure no duplicate same data unit, and also need make sure 9F74 and 9F69 is in the last returned record.

Note2: if qUICS and standard Debit/Credit use the same crypto version and algorithm identifier, then issuer application data can reuse 9200 or 9201 packets, or else, qUICS use 9207 and 9208 packets.

9.5.5 DGI 9209: Issuer Application Data (Offline Pre-authorization IDD option)

Tag	Data Element	Length
9F10	Issuer application data	Var.

Notes: This packet is optional, when this packet exists; this packet is only used in contactless low-value extended application offline pre-authorization transaction. If this packet does not exist, then offline pre-authorization IDD option is the same as standard qUICS, refer to 9207 packet.

Appendix A (Normative Appendix)

UICS Personalization Requirements Based on *EMV Card Personalization Specification*

This appendix describes a complete subset of EMV Card Personalization Specification for personalization, which is sufficient for personalization of issued products in specified market. Please contact UnionPay regional representative to discuss whether this appendix is appropriate for your market.

A.1 Data preparation

In the data preparation process, personalization data index (PDI) data is created and used to indicate personalization process, at the same time application data used for personalization of IC card application is also created. Data preparation process created data must be securely transported to the personalization device itself (unless the data is created in the personalization device HSM). In the data preparation process, any secret data created remotely by the personalization device must be encrypted prior to transmission, and all remotely generated data files must have message authentication code (MAC) protection prior to transmission.

Data preparation process includes the following steps:

- 1) Collecting personalization data from issuer source.
- 2) Grouping personalization data into data packets
- 3) Encrypt and calculate MAC as needed for personalization data.
- 4) Create personalization device commands.
- 5) Create data that needs to be recorded for application.
- 6) Create input file to be loaded into personalization device.

Personalization data fall into three categories:

- Master key and data of issuer
- Application key and certificate
- Application data

A.1.1 Issuer master key and data

Personalization could not proceed prior to the creation of issuer master key and other special data. Master key is used in two ways: support secure transport of personalization data, and create application level data for personalizing applications. Some data can be used to manage the personalization process, and other data is loaded to the card during personalization.

One or more master keys that are used in the personalization process may also be used in other processes. In this circumstance, a method to import or export master keys is needed to permit sharing of corresponding data between processes.

Prior to personalization, KMCID, key version number, KEYDATA and corresponding related keys must be loaded onto the card. KMCID and key version number is used to access issuer personalization master key (KMC), thereby using derivation data (KEYDATA) to derive and obtain the unique static sub-key for the card.

The 6 byte KMCID (ex: IIN number right align left pad 'F' if insufficient bits) and chip serial number (unique to each card) has its lowest 4 bytes used as key derivation data and must be stored in tag 'CF'. This data is also used to construct the return value to INITIALIZE UPDATE command.

Table A-1 - Key derivation data (tag 'CF')

Data Element	Description	Length	Format
KEYDATA	Key derivation data: - KMCID (6 bytes) -Chip serial number (4 bytes)	10	Binary

A.1.2 Application Key and Certificate

In the data preparation process, application level symmetric DES encryption keys must be created. In most cases, this key is derived from the corresponding issuer master key.

If fast dynamic data authentication (fDDA) functionality is used for qUICS, then issuer must generate a RSA key pair, and through authentication by payment system certificate authority, and then generate issuer certificate. Issuer certificate must be stored in a record on the card.

Also, fDDA functionality requires the card to support RSA public key technology, and generate a card RSA key pair, which is used to generate a card certificate through issuer using issuer private key for authentication. Card certificate must also be stored in a record on the card.

A.1.3 Application data

Application data falls into two categories. One category is issuer related, data is usable all IC cards (such as issuer id), another category is the unique data for each IC card (such as the PAN for a debit/credit application).

Logically or physically related data elements are packaged and defined within the same data grouping id (DGI). For example, DES keys are packaged in one DGI, because their tags belong to the same record, all data in the same packet will be sent to the IC card application through an independent command.

IC card application uses DGI to determine how to process data packets received from personalization device. Data packets are defined in Section 6. Personalization

device resolves the input record and create a corresponding STORE DATA command for each data packet in data record.

DGI must be encoded in binary format with 2 bytes, followed immediately with a binary length byte.

A.2 Interface: Personalization Device to ICC

A.2.1 Key Management

Prior to personalization sharing KMC, KENC, KMAC, and KDEK must be loaded into IC card application by personalization device or entity. Personalization device must be able to find KMC through issuer id and version number in the personalization device stored keys. For a specified IC card, the KMC id can be obtained from the card.

Table A-2 Personalization keys

Key Name	User	Usage Method	Master level	Card level	Session level
Issuer master key	Issuer, card manufacturer, and personalization device	Card manufacturer uses KMC to derive card level keys (KENC, KMAC, KDEK) then stores them into card.	KMC		
		Used to establish session key, this session uses CBC mode to create cryptogram and encrypt sensitive data.		KENC	SKUENC
		Used to create session keys, this session key is used to calculate C-MAC for command processing.		KMAC	SKUMAC
		Used to establish session key, this session uses ECB mode to encrypt DES keys and other optional sensitive data.		KDEK	SKUDEK
Issuer key exchange key	Issuer and data preparation device	Protects offline PIN and other sensitive data transported between issuer and data preparation device.	KEKISS		

Key Name	User	Usage Method	Master level	Card level	Session level
Data encryption key/transport key	Data preparation device and personalization device	Protects offline PIN and other sensitive data that is transported between preparation device and personalization device. Special data transport keys can be used in the following circumstances: PEK/TK - PIN encryption key used to protect PIN value. KEK/TK - Key exchange key used to protect DES key.	DEK / TK		
MACkey	Data preparation device provides to personalization device	Used to ensure integrity of application data provided to personalization device in personalization data file.	MACkey	N/A	N/A

KMC is used to create card independent static sub-key. Static sub-key is used to create the session key for IC card application communications (see A.4.2). Personalization keys are summarized in table A-2.

A.2.2 Processing Flow (commands)

This section describes the processing flow for data between interfaces. Assuming dual interface card is personalized through contact interface, personalization device enabled IC card (reset bit), and IC card responds with reset bit response data (ATR). Prior to contactless card sending SELECT command, it must be enabled and enter (ISO14443-4) protocol ready state. Type A card must receive selection response (ATS) or Type B card must receive "ATTRIB command" response, to ensure card is enabled.

Figure 3 summarizes the commands and response flow that must appear between personalization device and IC card.

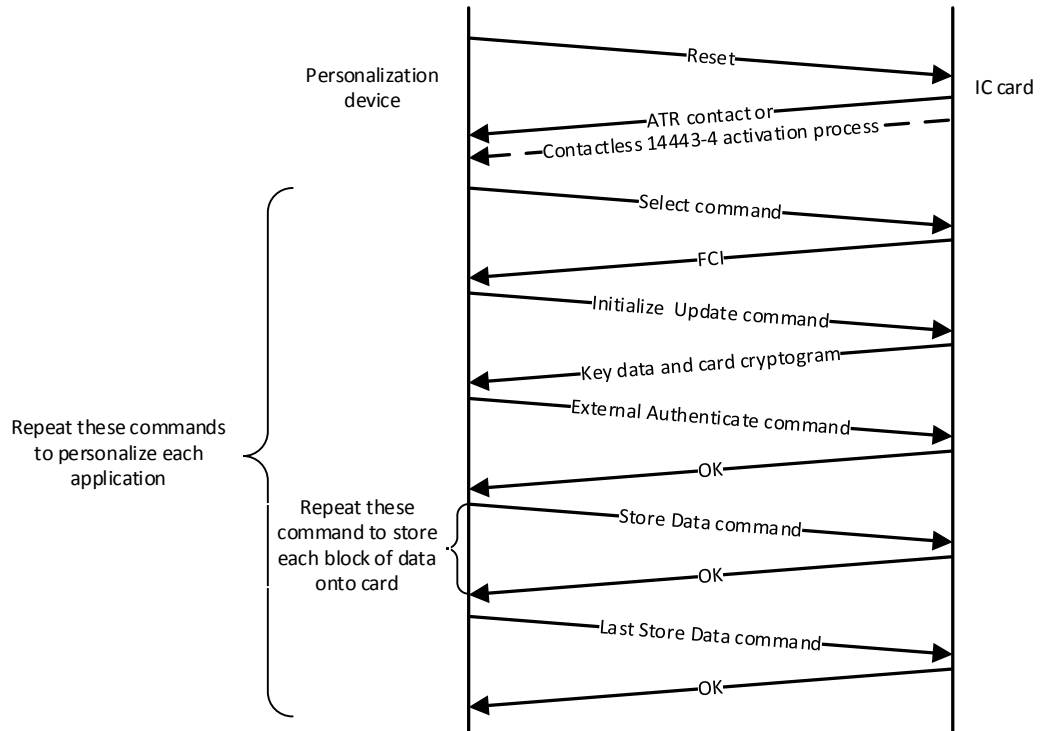


Figure 3 Personalization command flow

A.2.2.1 SELECT Command

SELECT command is used to select IC card application that needs personalization. Application selection is described in *EMV book 1*. For each IC card application that needs personalization, this command is sent only once.

A.2.2.2 INITIALIZE UPDATE Command

INITIALIZE UPDATE command is the first command sent by IC card personalization device after selecting the application. INITIALIZE UPDATE command is used to establish the secure session channel used in the personalization process. INITIALIZE UPDATE command exchanges the data used for mutual authentication, used to determine KMC id and version number, data used to derive KENC, KMAC, and KDEK are also returned through this command.

For each IC card application that needs personalization, INITIALIZE UPDATE command is sent only once.

INITIALIZE UPDATE command must follow the encoding in Table A-3. Terminal random number (RTERM) is generated by personalization device.

Table A-3 - INITIALIZE UPDATE command encoding

Field	Value	Length
CLA	'80'	1

Field	Value	Length
INS	'50'	1
P1	'00'	1
P2	'00'	1
Lc	'08'	1
Terminal random number (RTERM)	Random number used for host and card cryptogram calculations	8
Le	'00'	1

Table A-4 shows a successful response to INITIALIZE UPDATE command. INITIALIZE UPDATE command has no special status code, and all status codes for this command are defined in [ISO7816].

Table A-4 - INITIALIZE UPDATE command response data

Data Field	Length
KEYDATA (see table A-5)	10
Master key (KMC) version number	1
Secure channel protocol id (ALGSCP='02')	1
Counter	2
Card random number (RCARD)	6
Card cryptogram	8
SW1 SW2	2

Data Field		Length
Requirement A.1	Initial KEYDATA is provided to each IC card application as a data element. Its initial value is encoded according to table A-5.	

Table A-5 - KEYDATA initial content

Data Field	Length	Format
KMC id (Ex. IIN/BIN right align, if insufficient bits then left pad 1111b)	6	BCD
Chip serial number (CSN)	4	Binary

Requirement A.2 KMC id is part of INITIALIZE UPDATE command response data which is useful for determining issuer KMC.

Requirement A.3 The first 6 bytes of KEYDATA returned by INITIALIZE UPDATE command is used to distinguish master key (KMC) for creating secure data. The KEYDATA lowest 6 valid bytes are used as key derivation data. Personalization device must use KMC and KEYDATA for derivation to obtain the KENC, KMAC, and KDEK for each IC card as defined in Section A.3.1. These keys must be placed in IC card prior to the beginning personalization process.

Requirement A.4 Subsequent processing must use INITIALIZE UPDATE command response secure channel protocol (ALGSCP) id to determine how to create MAC and when to create session key.

Requirement A.5 Personalization device must generate card cryptogram and compare with INITIALIZE CUPDATE command returned card cryptogram to verify the validity of INITIALIZE UPDATE command returned card cryptogram. Card cryptogram is a MAC calculated through A.4.3.1 described method and SKUENC method, calculated MAC data variable y due to ALGSCP, calculated MAC data = terminal random number (8 bytes) || counter (2 bytes) || card random number (6 bytes). If card cryptogram verification fails, personalization processing is terminated

A.2.2.3 EXTERNAL AUTHENTICATE Command

EXTERNAL AUTHENTICATE command immediately follows INITIALIZE UPDATE command and is used by IC card application to verify the personalization device.

For each IC card application that needs personalization, EXTERNAL AUTHENTICATE command is sent only once.

Requirement A.6 EXTERNAL AUTHENTICATE command must follow encoding per table A-6, with terminal cryptogram calculated by personalization device.

Table A-6 -EXTERNAL AUTHENTICATE command encoding

Field	Value	Length
CLA	'84'	1
INS	'82'	1
P1	Security level -SECLEV = '00' Also, can select to support for security level 01 or 03 (see <i>EMV Card Personalization Specification</i>).	1
P2	'00'	1
Lc	'10'	1
Data	Host cryptogram	8
C-MAC	Command message authentication code	8

Requirement A.7 EXTERNAL AUTHENTICATE command return only contains status code: SW1 SW2

Table A-7 lists out possible returned status codes apart from those in [ISO 7816].

Table A-7 -EXTERNAL AUTHENTICATE command status codes

SW1	SW2	Meaning
'69'	'82'	MAC verification failure
'63'	'00'	Authentication of host cryptogram failed
'69'	'85'	Usage conditions not met

Subsequent commands received by IC card application will not include any security data, since there is no C-MAC, the entire command data is not encrypted.

Host cryptogram follows description in Section A.4.3.1 to use SKUENC to calculate a MAC. Data to calculate MAC = counter (2 bytes) || card random number (6 bytes) || terminal random number (8 bytes). IC card must calculate a host cryptogram, through comparison with value received from command data field the validity of host cryptogram can be verified.

C-MAC must be calculated from personalization device, and verified by IC card according to described method in Section A.4.3.2.

If EXTERNAL AUTHENTICATE command succeeds, serial counter increments by 1, and subsequently processes one or more STORE DATA commands.

Note: EXTERNAL AUTHENTICATE command has no response data to return

A.2.2.4 STORE DATA Command

STORE DATA command is used to personalize all applications. Each data packet corresponds to one STORE DATA command. STORE DATA command must establish a secure channel. The DGI used by STORE DATA command is determined based on input record data. The ENC field in input record lists data packet ids that must be encrypted.

Personalized IC card application STORE DATA commands must follow encodings shown in table A-8.

Table A-8 -STORE DATA command encoding

Field	Value	Length
CLA	'80' or '84' (if security levels '01' or '03' are supported, see <i>EMV Card Personalization Specification</i>)	1
INS	'E2'	1
P1	Highest bit indicates whether it is the last STORE DATA command (all other bits configuration shall be ignored by card application).	1
P2	Sequence no. - This byte can be ignored by card application	1
Lc	Command data field length	1
DGI	ID of data that must be stored	2
Length	Data packet length	1
Data	Data that must be stored	Var.

Table A-9 lists out possible returned status codes apart from those in [ISO 7816].

Table A-9 -STORE DATA command returned status codes

SW1	SW2	Meaning
'6A'	'88'	Related data not found (DGI not recognized)
'6A'	'80'	Data field parameter error

If DGI is listed as requiring encryption (ex., DGI '8000'), as described in the following section, data must be encrypted prior to being sent to IC card. Only the data part of the data packet is encrypted (DGI and length fields are not encrypted).

Personalization device uses session key SKUDEK to encrypt the data. Encryption must use ECB mode. This mode is described in Section A.4.4.1 3DES (ex. Requirement A.34) and is used to encrypt each packet.

Personalization device notifies IC card whether it is the last STORE DATA command by setting the 8th bit of P1 ('80'). So long as the secure channel is established, prior to directly receiving the STORE DATA command with P1 bit 8 set to 1, the card shall continue to accept STORE DATA commands. After the card receives the last STORE DATA command, subsequent STORE DATA commands shall return non-'9000' status words, ex. SW'6985' - usage requirements not met.

A.2.3 Command Response

Responses for all commands, regardless of whether processing succeeded, are all included and only included 2 byte status code: SW1 and SW2. SW1 and SW2 are both defined in [ISO7816] as 1 byte long.

Beside the special actions defined in this Appendix, when personalization device receives SW1 SW2 not equal to '9000', '6A88', '61xx', or '67xx', if could not recover, then personalization process shall be terminated.

A.3 IC Card Personalization Process

Prior to IC card personalization processing, there is a preparation or pre-personalization process. This preparation process is described here, to facilitate creating IC card needed data prior to personalization.

A.3.1 Preparation Prior to Personalization (pre-personalization)

Prior to personalization, IC card must be enabled/activated, basic applications downloaded, and file and data structures established. Also, some specified data must be stored into the IC card. In some circumstances, these data are appropriate for the entire card (ex. KMCID). In some circumstances, these data are only appropriate for a single application (ex. AID).

Requirement A.8	Unless the card supports dynamic creation of files and records and initializing their space to binary zeros, then the IC card application must have files created and storage space allocated according to Specification described requirements, and initialized to binary zeros.
Requirement A.9	All applications must be selected through AID.
Requirement A.10	If an application's FCI does not require personalization then this application's FCI must be created prior to personalization.

Requirement A.11	KEYDATA must be set according to Table A-5 requirements, and KEYDATA is composed of KMCID and chip serial number (CSN). KMCID is the personalization master key ID provided by issuer or personalization executing party, KMCID length is 6 bytes. CSN is the right-most 4 bytes of the card physical id.
Requirement A.12	Personalization master key (KMC) version number is used to generate initial personalization keys (KENC, KMAC, and KDEC) for each application, with this version number stored in the 'CF' tag of the IC card (see table A-2).
Requirement A.13	A derived key (KENC) shall be generated for each IC card and stored into the card. This key is used to calculate card cryptograms and verify host cryptograms. KENC key is a 16 byte DES key (112 bit data plus 16 but verification value). KENC key is derived through the following method: $KENC := DES3(KMC)[KEYDATA \text{ lowest } 6 \text{ bytes} 'F0' '01'] DES3(KMC)[KEYDATA \text{ lowest } 6 \text{ bytes} '0F' '01']$.
Requirement A.14	A derived key (KMAC) shall be generated for each IC card and stored into the card which is used to verify external authentication command's C-MAC value. KMAC key is a 16 byte DES key (112 bit data plus 16 but verification value). KMAC key is derived through the following method: $KMAC := DES3(KMC)[KEYDATA \text{ lowest } 6 \text{ bytes} 'F0' '02'] DES3(KMC)[KEYDATA \text{ lowest } 6 \text{ bytes} '0F' '02']$.
Requirement A.15	A derived key (KDEC) shall be generated for each IC card and stored into the card which is used to decrypt in ECB mode the encrypted data received through STORE DATA command. KDEC key is a 16 byte DES key (112 bit data plus 16 but verification value). KDEC key is derived through the following method: $KDEC := DES3(KMC)[KEYDATA \text{ lowest } 6 \text{ bytes} 'F0' '03'] DES3(KMC)[KEYDATA \text{ lowest } 6 \text{ bytes} '0F' '03']$.
Requirement A.16	Counter returned through INITIALIZE UPDATE command must be initialized to '0000'.

A.3.2 Personalization Requirements

A.3.2.1 IC Card Requirements

Requirement A.17 Application to be personalized must be already installed on the card and can be selected through the application selection process defined in [EMV Book 1] Part II.

A.3.2.2 Command Support

Requirement A.18 All IC cards that support this Specification must support the personalization commands described in Section A.2:

- SELECT
- INITIALIZE UPDATE
- EXTERNAL AUTHENTICATE
- STORE DATA

Requirement A.19 Each IC card must maintain a counter that can be returned through INITIALIZE UPDATE command. This command must be incremented by one with each successful EXTERNAL AUTHENTICATE command.

Requirement A.20 If STORE DATA command has DGI that the IC card application does not recognize, then return status word '6A88'.

A.3.2.3 Security Data

All application EXTERNAL AUTHENTICATE command requires security messages. (See Section A.2.2.3)

Requirement A.21 EXTERNAL AUTHENTICATE command must include an 8-byte C-MAC, and IC card must first verify it prior to accepting commands. If C-MAC verification fails, IC card shall reject command and also return status word = '6982' and close the secure session channel.

Requirement A.22 In order to verify C-MAC, IC card must calculate a C-MAC and compare with the C-MAC received in the command. C-MAC must be calculated following the description in A.4.3.2.

Requirement A.23 Card application must be able to decrypt data defined in Section A.4.5.

A.4 Personalization Encryption

A.4.1 Security Principles

- STORE DATA command is only executed with secure session channel established.
- Card must support the subset of secure channel protocol defined in *EMV Card Personalization Specification*. This subset has complete definition in this document.
- DES key uses odd verification.
- All personalization DES keys are double length keys (16 bytes)
- All MACs are 8 bytes in length.

A.4.2 Session Keys

DES session keys are generated each time secure channel is established, with a max of three session keys generated, ie. SKUENC, SKUMAC, and SKUDEK.

Requirement A.24 All encryption, decryption, and MAC calculations in commands sent to IC card must use session keys (SKUENC, SKUMAC, and SKUDEK).

Requirement A.25 Session keys SKUENC, SKUMAC, and SKUDEK must be calculated using base keys KENC, KMAC, and KDEC according to 3DEC algorithm described in Section A.4.4.2.

Session keys must be calculated according to CBC mode described in Section A.4.4.2 and data in Table A-10. No padding is necessary prior to encryption. The 16 byte derivation data after encryption yields a 16 byte double length key.

Table A-10 - Session key derivation data

Key	IC Card Key	Derivation Data
SKUENC	KENC	'0182' Counter '000000000000000000000000'
SKUMAC	KMAC	'0101' Counter '000000000000000000000000'
SKUDEK	KDEK	'0181' Counter '000000000000000000000000'

Session keys must be calculated each time IC card application processes INITIALIZE UPDATE command using IC card provided counter. For INITIALIZE UPDATE command description, please see Section A.2.2.2.

These session keys would be used in the cryptogram calculations over the entire IC card application personalization process, until the last STORE DATA command ends.

A.4.3 MAC

The goal of creating MAC for the personalization process is as follows:

In the IC card personalization process (INITIALIZE UPDATE command and EXTERNAL AUTHENTICATE command), IC card returns a MAC (card cryptogram) to the personalization device, and personalization device sends a MAC (host cryptogram) to the IC card. IC card and personalization device use these cryptograms to authenticate each other. To calculate these MAC processes, please see Section A.4.3.1.

When personalization device sends external authentication commands, there is a C-MAC to be delivered to the card. For the calculation process of this MAC, please see Section A.4.3.2.

A.4.3.1 Personalization of Encrypted MACs

Requirement A.26	To calculate MAC input data, pad a byte '80' at first right, and on the right side pad at most 7 '00' bytes, until the result is an integer multiple of 8 bytes. Please see [ISO 9797] defined padding method 2.
Requirement A.27	Full 3DES MAC is MAC algorithm 1 defined in [ISO9797], uses output conversion 1, no truncation, and uses 3DEC in place of block encryption.
Requirement A.28	All 64 bit final calculation results use the MAC in personalization cryptogram.
Requirement A.29	Cryptogram verification process must use the same method and parameter (key) to calculate a MAC and compare against received cryptogram.

A.4.3.2 Secure Message C-MAC

Requirement A.30 Prior to accepting other commands, IC card shall receive and verify an external authenticate command generated by personalization device (see Section A.2.2.3), this command includes an 8-byte C-MAC. If verification of C-MAC fails, IC card shall reject the command, return status word '6982', and close the secure channel.

Note: To differentiate from the MAC functionality defined in A.4.3.1, C-MAC is used as the function name for generation of secure MAC data and as the name of the secure MAC data.

Requirement A.31 C-MAC calculation method is as follows:

Connect command head (CLA INS P1 P2 Lc), as shown in Table A-6, and connect command data fields (not including C-MAC itself).

C-MAC calculation is as follows, use SKUMAC key.

Requirement A.32 Process of calculating C-MAC is executing single DES calculation plus the last 3DES MAC calculation method, this method is defined as MAC algorithm 3 in [ISO 9797], uses output conversion 3, no truncation, and uses DES in place of block encryption. This algorithm is also referred to as "Retail MAC", as described in Figure 4.

Note: In this figure,
 DES represents single DEC encryption,
 DES-1 represents single DES decryption,
 SK_L indicates the leftmost bytes of the session key,
 SK_R indicates the rightmost bytes of the session key

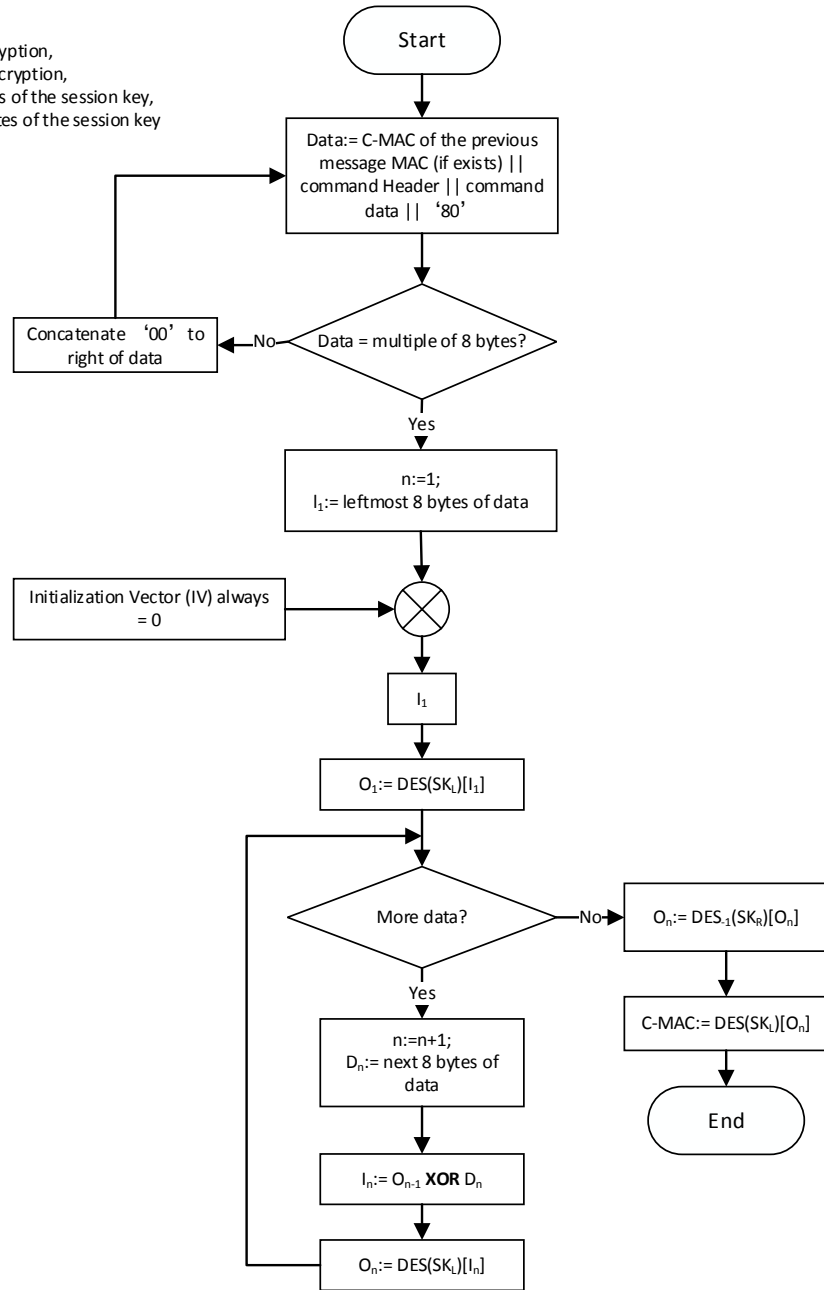


Figure 4 C-MAC calculation

Personalization device and IC card both require calculation of C-MAC. IC cards verify C-MAC by comparing self-generated C-MAC with the C-MAC received in command.

A.4.4 Encryption

This section describes encryption of sensitive data in the personalization process. Note that post-personalization encryption is not appropriate in MSD applications.

A.4.4.1 ECB Mode Encryption

Requirement A.33 Personalization device must encrypt keys, and use 3DES in ECB mode and SKUDEK as session key.

Requirement A.34 Use the 3DES ECB mode defined in [ISO 10116].

A.4.4.2 CBC Mode Encryption

Data encryption must use the 3DES CBC mode defined in [ISO 10116], initial vector is '00 00 00 00 00 00 00 00'.

A.4.5 Decryption

IC card shall decrypt data prior to storing sensitive data for future use. This section describes decryption of sensitive data in the personalization process.

A.4.5.1 Decryption

IC card applications use SKUDEK to decrypt the encrypted data packets, using the 3DES CBC mode described in [ISO 10116].

A.4.6 3DES Calculation

3DES uses compound DES encryption and decryption operations. The 3DES appropriate for this Specification is the content defined in key option 2 of [ISO 18033-3].

Appendix B

(Informative Appendix)

Personalization Examples

This appendix provides the personalization log that demonstrates the EMV CPS subset defined above, used to support PPSE, UICS, and qUICS cards. Card applications (PPSE, UICS, and qUICS) can be accessed through a UICS AID. Card shall include the following applications, and applications are prepared for personalization:

PPSE (AID '32 50 41 59 2E 53 59 53 2E 44 44 46 30 31')

UICS and qUICS (if supported) (AID 'A0 00 00 03 33 01 01')

Note: The personalization log here is an example, and does not contain all possible options and data elements.

B.1 Contactless Payment System Environment (PPSE)

Select PPSE for Personalization

Command --> '00 A4 04 00 0E'

'32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 00'

<- Return '6F 10 84 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31'

Establish Secure Channel (SCP-02)

Command --> '80 50 00 00 08' INITIALIZE UPDATE Command

'00 22 44 66 88 AA CC EE' '00'

<- Return '00 00 00 00 00 00 00 00 00 00 01 02 00 01 CC CC CC CC CC CC 08 D0 03 00 E7 AE BA 21'

Command --> '84 82 00 00 10' EXTERNAL AUTHENTICATE
Command

'30 61 1C F0 07 DD 70 2B

'03 07 87 EA 54 C0 D6 AD'

STORE DATA

Command --> '80 E2 80 00 20' Note that P1 = '80'

'91 02 1D' SELECT Return

'A5 1B'

'BF 0C 18'

'61 16'

'4F 07 A0 00 00 03 33 01 01'

'50 0B 56 49 53 41 20 43 52 45 44 49 54'

B.2 Payment Application (qUICS)

The following example is a personalization process of a qUICS application that includes the following functionality:

- Micropayment and CTTA(qUICS)
- fDDA version 01(qUICS)

If contact UICS is supported, there is an additional DGIs and data element that must be personalized. UICS personalization example is as shown below, but notes the example does not include all possible UICS data.

Select Payment Applications for Personalization

Command --> '00 A4 04 00 07'

' A0 00 00 03 33 01 01'

'00'

<- Return'6F 09 84 07 A0 00 00 03 33 01 01'

Establish Secure Channel (SCP-02)

Command --> '80 50 00 00 08'

INITIALIZE UPDATE Command

'00 22 44 66 88 AA CC EE'

'00'

<- Return'00 00 00 00 00 00 00 00 00 00 01 02 00 02 CC CC CC CC CC CC D9
1E 0E 08 23 4C 3B E9'

Command --> '84 82 00 00 10'
Command

EXTERNAL AUTHENTICATE

'02 DD 99 D5 FF 9C D4 9C'

'02 7D F6 D2 29 AC 4F AA'

STORE DATA

Command --> '80 E2 00 01 1B'

'92 07 18' QUICS GPO Return Data

'82 02' QUICS AIP

'20 00'

'94 08' QUICS AFL

'18 01 01 01 10 01 02 00'

'9F 10 08' Issuer Application Data (CVN 17)⁴

'07 01 17 03 00 00 00 01'

Command --> '80 E2 00 01 15'

'91 04 12' UICS GPO Return (contact) Data

'82 02' UICS AIP

'7C 00'

'94 0C' UICS AFL

'08 02 02 00 10 01 03 00 18 01 02 01'

Command --> '80 E2 00 01 2A'

'01 01 27' SFI 1 RECORD 1

'70 25'

'57 13' Magnetic track 2 equivalent data

'62 61 73 90 01 01 00 10 D1 01 22 01 01 23 49 99 99 99 1F'

'5F 20 0D' Cardholder Name

'45 58 41 4D 50 4C 45 2F 50 45 52 53 4F'

Command --> '80 E2 00 01 42'

'01 02 3F' SFI 1 RECORD 2

'70 3D'

'57 10' Magnetic track 2 equivalent data

'62 61 73 90 01 01 00 10 D1 01 22 01 01 23 45 56'

'5F 20 0D' Cardholder Name

'45 58 41 4D 50 4C 45 2F 50 45 52 53 4F'

'9F 1F 18' Magnetic Track 1 Self-defined Data

'30 31 32 33 34 30 30 30 30 30 30 30 30 35 35 36 30 30 30 30 30 30'

Command --> '80 E2 00 01 99'

'02 01 96' SFI 2 RECORD 1

'70 81 93'

'90 81 90' Issuer public key certificate

'61 B5 F7 4B EB 13 5D 31 B7 3A 0B 17 BF D8 C5 A3 28 45 CC 08 0F 34 88 DA
B1 95 0C 6C 55 89 C3 13 37 B5 30 08 88 B1 67 C9 F2 83 B7 1C D2 58 02 4E 32
FC 3E 39 AC F5 60 1C 9E 99 25 FC 61 58 9D C3 38 FF B4 5C D7 07 43 7D 5F
77 87 2F 46 EB AB DB 4C 23 01 02 69 DF DE 0B E8 A9 E6 16 53 8C 8D E5 57
DC 02 AA B3 5B 28 CA DD 38 20 7F A1 16 68 14 FD 19 E8 03 77 5D 0B A3 EA
3B C2 E4 75 AD 40 DF AD 2E CA 82 17 E0 CD EF 86 6B 28 44 A4 6E 6C 6B'

Command --> '80 E2 00 01 C0'

'02 02 BD' SFI 2 Record 2

'70 81 BA'

'8F 01' CA public key index

'95'

'92 14' Issuer public key remainder

'A2 EF A5 CB F0 2C C4 7D 47 83 3B B7 B2 7E CC 69 62 38 5A 4B'

'9F 32 01' Issuer Public Key Exponent

'03'

'9F 46 81 80' IC card public key certificate

'A7 2A 76 A9 DB 48 2E 8B 01 66 3C FE CD 4C BA 96 E8 71 E9 1B F9 CA 74 C0
 19 09 D1 00 6A DC 31 4B 6C 17 BD 93 A0 AE F2 12 65 40 A9 9A 24 97 6A 25
 9A 82 64 F9 BD 00 7F 24 45 86 55 FD 3F 72 F3 AF 0F A7 8F DB 1B 0F 0C C8
 67 E0 6D 18 3E AA E5 5D 08 D3 36 49 D0 48 86 F1 DD 94 19 CA AC E8 45 31
 23 DE 13 FC B1 5C 34 8B CD F7 70 54 C6 85 E4 C8 B5 33 3E AF 95 5C 6C D4
 73 FF AD E8 59 36 23 E4'

'9F 47 01' IC card public key index

'03'

'9F 48 0A' IC card public key remainder

'BA A5 EB 09 E4 4B 5D 59 33 4F'

'9F 69 05' Card authentication related data

'01 00 00 00 00'

Command --> '80 E2 00 01 89'

'02 03 86' SFI 2 Record 3

'70 81 83'

'93 81 80' Static Data Signature

'78 82 CA 7E CA 13 09 9D C2 15 E6 35 60 B2 E3 19 84 74 27 C5 59 BD A2 2A
 FB 7C 77 63 4D 6E A2 32 60 03 01 9C 85 9D D8 A6 13 4A 2F 13 7D 53 FA 53
 B2 24 70 3D 55 40 8F 3E 15 BB 3B 01 B7 64 DB 15 16 DA 18 1C 0C 70 37 D9
 D2 10 FC F6 D7 F3 97 11 99 D1 02 C2 D1 24 0B 86 1D E3 B6 CE 13 D4 02 2E
 54 B9 38 0B 3B FF EB 89 98 FF CE 65 E5 CF AB 82 3F 29 F3 9A E9 7C 62 27
 A2 88 BA DC 43 9C 8B FF'

Command --> '80 E2 00 01 15'

'03 01 12' SFI 3 Record 1

'70 10'

'5A 08' Primary Application Number PAN

'47 61 73 90 01 01 00 10'

'5F 24 03' Application Expiration Date

'10 12 31'

Command --> '80 E2 00 01 6E'

'03 02 6B' SFI 3 Record 2

'70 69'

'5F 30 02' Service Code

'02 01'

'5F 34 01' PAN Serial No

01'

'8C 15' CDOL1

'9F 02 06 9F 03 06 9F 1A 02 95 05 5F 2A 02 9A 03 9C 01 9F 37 04'

'8D 17' CDOL2

'8A 02 9F 02 06 9F 03 06 9F 1A 02 95 05 5F 2A 02 9A 03 9C 01 9F 37 04'

'8E 0C' CVM List

'00 00 00 00 00 00 00 00 02 03 1F 00'

'9F 08 '02' Application Version No

'00 8C'

'9F 0D 05' Issuer Action Code - Default

'F0 20 04 00 00'

'9F 0E 05' Issuer Action Code - Reject

'00 50 88 00 00'

'9F 0F 05' Issuer Action Code - Online

'F0 20 04 98 00'

'9F 42 02' Application Currency Code

'08 40'

Command --> '80 E2 00 01 0D'

'92 00 0A' Issuer Application Data

'9F 10 07'

'06 01 0A 03 00 00 00'

Command --> '80 E2 00 01 72'

'0E 01 6F' Internal Data

'57 10'	Magnetic Track 2 Equivalent Data
'47 61 73 90 01 01 00 10 D1 01 22 01 01 23 45 56'	
'5F 34 01'	PAN Serial No
'01'	
'9F 51 02'	Application Currency Code
'08 40'	
'9F 52 04'	Application Default Action
'82 40 00 00'	
'9F 53 01'	Consecutive Offline Transaction Counter
'05'	
'9F 54 06'	Cumulative Offline Transaction Amount
'00 00 00 01 00 00'	
'9F 56 01'	Issuer Authentication Indicator
'80'	
'9F 58 01'	Consecutive Offline Transaction Lower Limit
'03'	
'9F 59 01'	Consecutive Offline Transaction Upper Limit
'07'	
'9F 5C 06'	Cumulative Offline Transaction Amount Upper Limit
'00 00 00 02 00 00'	
'9F 68 04'	Card Additional Processing
'4C 00 00 00'	
'9F 6C 02'	Card Transaction Qualifiers
'20 00'	
'9F 77 06'	UICS Micropayment Balance Upper Limit
'00 00 00 00 40 00'	
'9F 78 06'	UICS Micropayment Single Transaction Limit
'00 00 00 00 25 00'	
'9F 79 06'	UICS Micropayment Available Amount
'00 00 00 00 15 00'	
Command --> '80 E2 00 01 33'	
'80 00 30'	DES Key

'D6 16 01 7F BF 63 AC E7 7A 37 EB 7A EE 92 1F A5'

'25 FC 6E C6 D5 C2 BF F7 E7 29 4B 67 26 F2 48 6B'

'98 AE 3F 24 26 25 FC FA FC CE 0A 05 20 38 CF E6'

Command --> '80 E2 00 01 0C'

'90 00 09' DES Key Verification Value

'53 8B 0E'

'A8 B8 EE'

'E3 BA A0'

Command --> '80 E2 00 01 3B'

'82 01 38' IC CARD KEY(CRT)

'AF 30 C1 B4 78 37 B8 60 82 72 2A FB 00 F1 92 32 02 25 7D 6F 78 B7 E2 CF 1B
C0 2C AF 9A 7A 2B 5B CE 0B CA AF 87 01 4E FB 9A A0 80 CD A4 1B 6B 62
B2 EC 54 DC 43 AC F5 7F'

Command --> '80 E2 00 01 3B'

'82 02 38' IC CARD KEY(CRT)

'13 DB 1E C6 34 9B B7 DF 02 F0 77 73 C4 B1 31 EA B9 D3 DD 10 72 9F B5 7D
2F E8 A1 B4 C0 83 91 AF 7C B6 62 2B CB A8 35 B6 80 7A DB B9 38 07 41 E9
B2 EC 54 DC 43 AC F5 7F'

Command --> '80 E2 00 01 3B'

'82 03 38' IC CARD KEY(CRT)

'C2 67 74 EA E8 59 CC 93 03 64 5D A8 37 CA D1 37 39 FA 74 08 EC 25 FD 33
2C 23 45 D5 47 6C 00 86 6D 46 3A 66 D6 B4 C8 43 23 60 39 98 7C 98 D7 96 B2
EC 54 DC 43 AC F5 7F'

Command --> '80 E2 00 01 3B'

'82 04 38' IC CARD KEY(CRT)

'E6 7D B5 E5 75 76 80 18 45 11 D8 E5 B9 8B A6 5A 17 79 07 97 C8 88 08 14 E8
26 C1 44 71 56 49 E0 20 79 E9 F3 31 39 F7 83 9D F9 DB C1 6D B0 6D F9 B2 EC
54 DC 43 AC F5 7F'

Command --> '80 E2 00 01 3B'

'82 05 38' IC CARD KEY(CRT)

'62 08 BC 81 56 E9 65 FE 77 29 7B 08 03 A8 B3 02 C2 7F FD 2B 10 46 70 85 4F
37 A6 5A E4 2D 22 51 55 6C B3 86 07 4C 11 9F D6 47 D7 71 AC 17 8D 0C B2
EC 54 DC 43 AC F5 7F'

Command --> '80 E2 00 01 12'

'91 02 0F'	SELECT Return
'A5 0D'	
'50 0B' Application Tag	
'56 49 53 41 20 43 52 45 44 49 54'	
Command --> '80 E2 80 01 27'	
'91 03 24'	Contactless SELECT Return
'A5 22'	
'50 0B' Application Tag	
'56 49 53 41 20 43 52 45 44 49 54'	
'9F 38 0C' PDOL (CVN 17)	
'9F 66 04	
9F 02 06	
9F 37 04	
5F 2A 02'	

B.3 Response After Personalization

B.3.1 Select PPSE

```
Command --> '00 A4 04 00 0E'

'32 50 41 59 2E 53 59 53 2E 44 44 46 30 31'

'00'

<- Return'6F 2D'

'84 0E'

'32 50 41 59 2E 53 59 53 2E 44 44 46 30 31'

'A5 1B'

'BF 0C 18'

'61 16'

'4F 07 A0 00 00 03 33 01 01'

'50 0B 56 49 53 41 20 43 52 45 44 49 54'
```

B.3.2 Select Payment Application

```
Command --> '00 A4 04 00 07'
'A0 00 00 03 33 01 01
'00'
<- Return'6F 2D'
```

'84 07'

'A0 00 00 03 33 01 01'

'A5 22'

'50 0B'

'56 49 53 41 20 43 52 45 44 49 54'

'9F 38 0C'

'9F 66 04

9F 02 06

9F 37 04

5F 2A 02'